

NYU, Tandon School of Engineering

Bridge to CS

Homework #12

Programming Style

Pay special attention to the style of your code. Indent your code correctly, choose meaningful names for your variables, define constants where needed, choose the most appropriate control flow statements, and break down your solution into functions where appropriate.

Problem Description

Write a checkbook balancing program. This program will help the user keep track of their outgoing checks and allow the user to know the difference between what their bank account displays as their account balance and what it will be when all outstanding checks are cashed.

The program will read in, from the console, the following information for all checks that were not recorded as of the last time the checkbook was balanced:

- The number of each check (**int**)
- The amount of the check (**Money**)
- Whether or not the check has been cashed (1 or 0)

The check amount will be of type **Money** (as defined on page 662 in the textbook; Display 11.9). You should **copy the Money class from the provided .cpp file on Brightspace** into your program. The **Check** class should have accessor and mutator functions, constructors, and functions for both input and output of a check.

Store the user-input checks in an array, using the **Check** class as the element type.

Additional Requirements

In addition to reading in checks, the program should also act as a bank account. The program should:

- Read in the old account balance from the user.
- Read in all new deposits from the user.
- Store deposits (if desired) in a separate array.

This functionality may be implemented using **main** and helper functions and does not require an additional class.

Program Output

Your program should output:

- Two lists of checks:
 - Checks that have been cashed since the last time the checkbook was balanced
 - Checks that have not yet been cashed
- The sum of the cashed checks and the sum of the uncashed checks
- The total amount of deposits
- The new account balance according to the bank:

`Old balance + deposits - cashed checks`

- The actual account balance after all outstanding checks are cashed:

`Bank balance - uncashed checks`

- The amount by which the actual balance differs from the bank-reported balance

Extra Credit: Display both lists of checks in sorted order from lowest to highest check number.